

EV Challenge Team Report

1. Team Information

- **Team Name:** IITGRacing
- **Team Members:**
 - Ronak
 - Ryan Sequeira
 - Bedabrat Barma
 - K Aravindhnan
 - Aayush Kumar
- **Mentor Name & Affiliation:** Mohammed Mosaddique Nawaz, IIT Guwahati EEE Department Professor

2. FMU Details

- **FMU Created:** Yes
- **If Yes, Provide Details:**

A modular EV simulation model was created in Simulink and exported as a Functional Mock-up Unit (FMU) using the FMI Toolbox for co-simulation. The FMU simulates optimal velocity, energy usage and real-time motor torque control based on live feedback from CarMaker.
- **FMU Generation Method:**

The FMU was generated using MATLAB Simulink's FMI Export Toolbox, compliant with FMI 2.0 for Co-Simulation.

3. FMU Interface and Parameters

- **Number of Interfaces Added:** 3 interfaces:
 - Battery energy, distance travelled, and vehicle velocity (inputs)
 - V_opt (optimal speed, intermediate output)
 - Motor_torque_trg (motor torque demand)

- **Parameters Used in FMU:**

Parameter	Symbol	Value	Units
Air density	$\rho(\rho)$	1.225	kg/m ³
Drivetrain Efficiency	$\eta(\eta)$	0.9	-
Gravity	g	9.81	m/s ²
Frontal Area	A	2.48	m ²
Vehicle Mass	m	1400	kg
Rolling Resistance	C_{rr}	0.01	-
Drag Coefficient	C_d	0.31	-
Maximum Velocity	max_veh	16.667	m/s
Gear Ratio	G	6	-

4. Simulation Strategy

- **Approach to Choosing Optimal EV Speed:**

The simulation was structured to determine the optimal speed (V_{opt}) required to complete a **153 km** journey using a limited battery capacity (**90% SoC**). It uses physical forces (rolling resistance, aerodynamic drag, acceleration force) to **compute required motor torque** and energy usage dynamically, updating the motor torque demand in real-time.

Control Strategy:

- Calculate the **optimal speed** of the car based on available **Energy Remaining /m**.
- A **PID controller** compares actual and target velocities to **generate an acceleration signal**.
- This acceleration is then converted into **net force (F_{net})**, which is broken down into motor force requirements by adding the resistive forces.
- **Torque output** is adjusted via a gear ratio block and supplied to the motor.

Optimal Speed Model:

The Simulink model computes V_{opt} based on:

$$V_{opt} = \sqrt{\{ [(E/d) \cdot \eta - C_{rr} \cdot m \cdot g] / (0.5 \cdot \rho \cdot C_d \cdot A) \}} \quad - (1)$$

The V_{opt} value is constrained to not exceed the value of max_veh (16.667 m/s).

The Simulink model to compute the **Net Force** based on:

$$F_{net} = C_{rr} * m * g + 1/2 * \rho * C_d * A * veh_{actual}^2 + m * a \quad - (2)$$

- **Observations During Simulation Execution:**

- With a realistic drag coefficient and low rolling resistance, the vehicle achieves best efficiency around 14.5–15.8 m/s. At times, the actual velocity of the car does cross `max_veh` (Overshooting) and is immediately brought down right before reaching 22 m/s (80 kmph).
- When energy drops due to losses, optimal speed gradually reduces.
- Any NaN in `V_opt` appears due to invalid square root inputs (negative under-root), which was later rectified with saturation blocks.
- The total distance of **153 km** was covered in **9612.2 seconds (2.67 hours)**, with **19% SoC** remaining in the end.

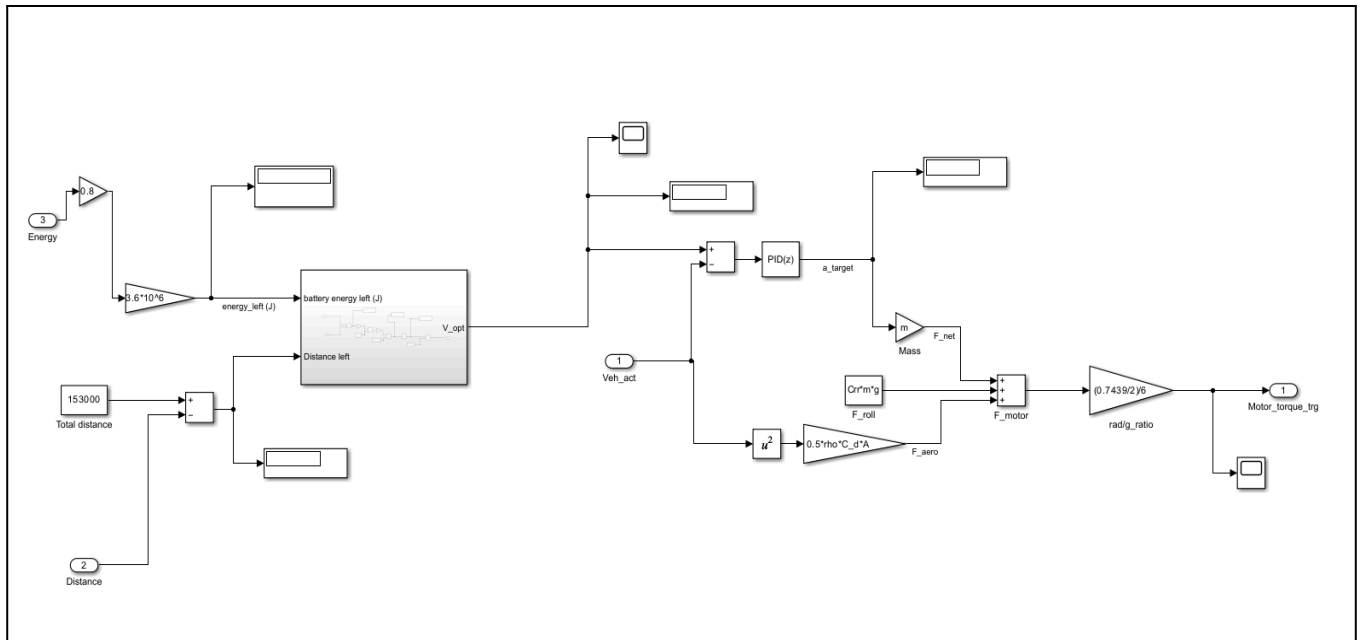
5. Team Experience and Learning

- **Team's Experience in This EV Challenge:**

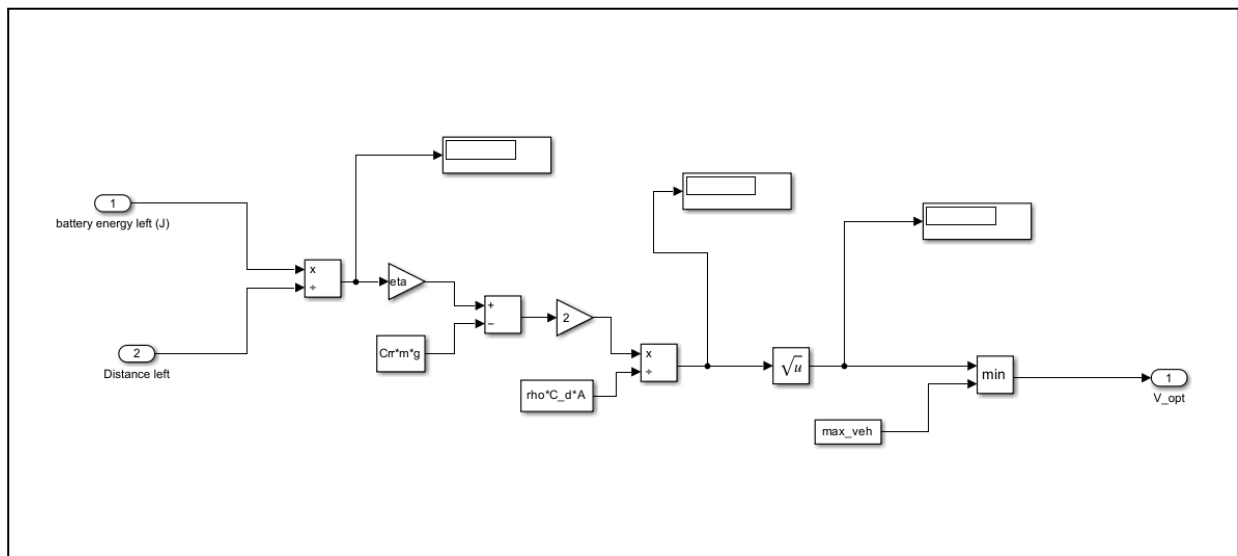
- We gained experience in developing FMU-compatible models and integrating physical equations within Simulink.
- Learned to model real-world energy constraints and translate them into controller logic.
- Understood the complexity behind balancing efficiency and speed in EVs.
- Explored drivetrain behavior, torque demands, and force interactions in detail.
- Improved our skills in data-driven simulation and debugging FMUs in test environments.

6. Optional Attachments

FMU Block Diagram:



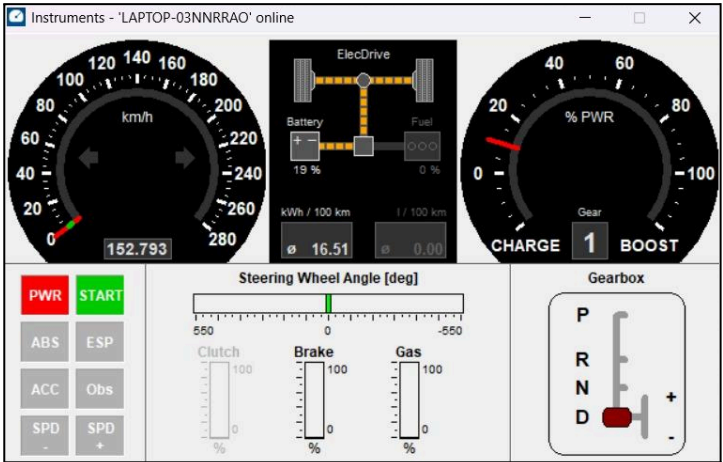
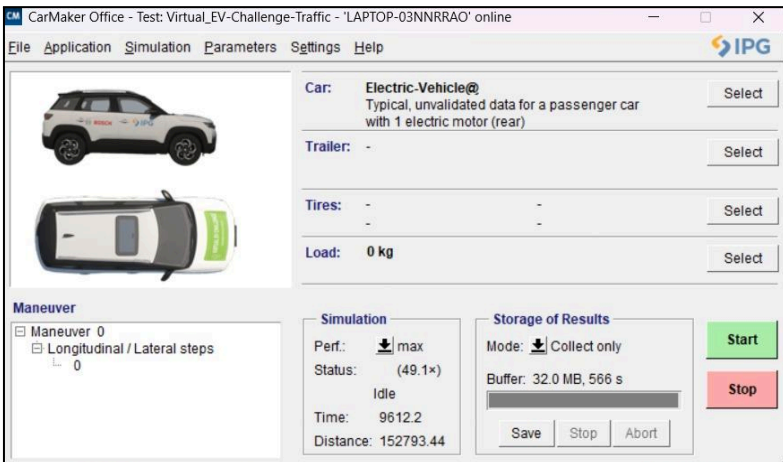
Optimum Velocity Subsystem:



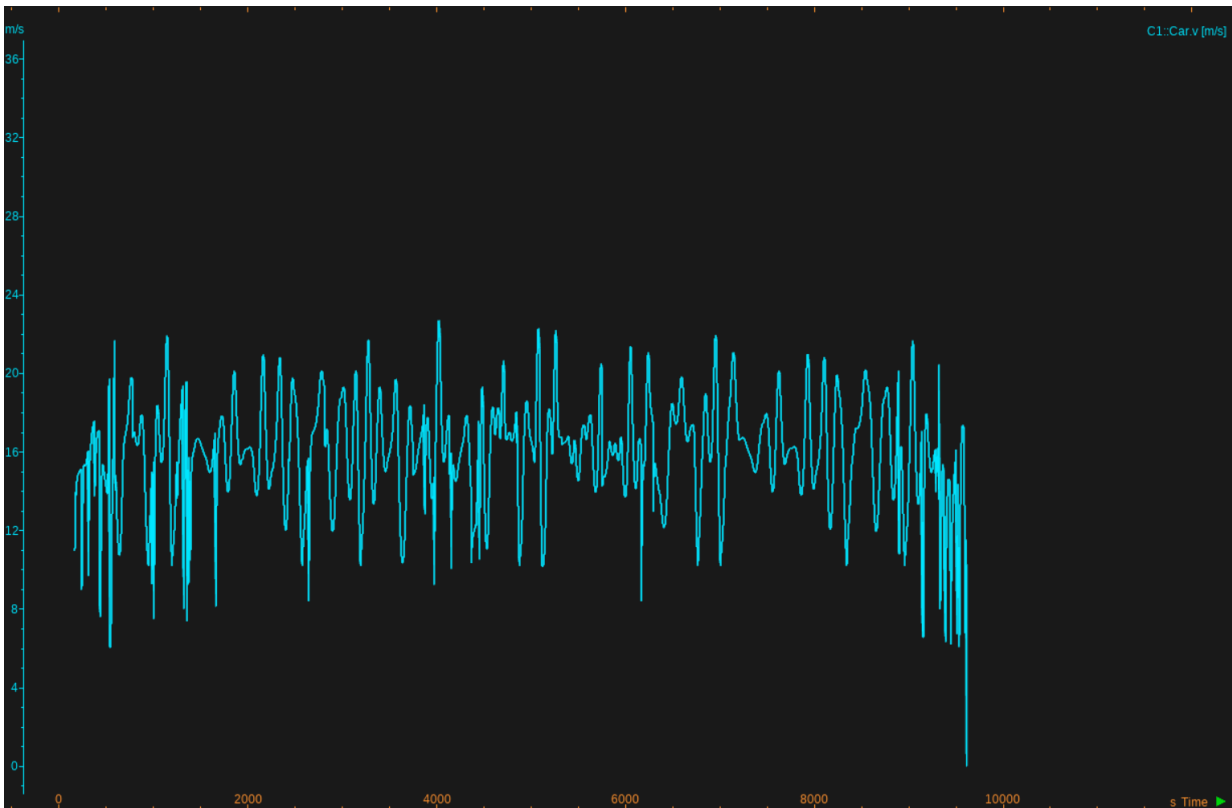
PID Controller Parameters:

Controller parameters	
Source:	internal
Proportional (P):	0.4
Integral (I):	0.000001 ☐ Use I*Ts (optimal for codegen)
Derivative (D):	0 ☐ Use externally sourced derivative
Filter coefficient (N):	100 ☒ Use filtered derivative

Simulation Results:



Velocity v/s Time:



Submitted on: 27 June 2025

Submitted by: Ryan Sequeira